Maduka Jayasinghe

Systems Software Engineer
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Open to Remote Work | Open to Relocation

Summary

Systems Software Engineer with 5+ years of experience developing high-performance software for network visibility, control, and network management systems (NMS). Strong expertise in low-level development using C, C++, Python, and Java, with hands-on experience in hardware-software co-design, deep packet inspection, and system performance optimization. Proven track record of leading engineering teams in delivering scalable computing systems, network monitoring infrastructure, and robust cross-platform integrations.

Key Achievements

- Led the end-to-end architecture and delivery of an enterprise-grade Network Monitoring System for ISP and enterprise clients, enabling real-time network visibility and monitoring.
- Conducted technical training and deployment workshops for the Paraqum product suite across Sri Lanka and Bangladesh to drive regional partner enablement.

Professional Experience

Senior Systems Software Engineer - II Apr. 2026 – Present
Paraqum Technologies Pvt. Ltd. *Sri Lanka*

- **NMS Product & Backend Engineering Lead:** Directed product strategy and backend development for an enterprise Network Monitoring System using C++ and Node.js, integrating SNMP, WMI, SSH, and Telnet protocols for real-time telemetry and device monitoring.
- **High-Throughput DPI Engine:** Developed core C/C++ backend modules leveraging DPDK for the deep packet inspection (DPI) engine, powering network analyzer and traffic shaper platforms to enable multi-gigabit traffic processing with minimal latency.
- **Network Configuration Management Platform:** Designed and developed a network configuration management platform to automate provisioning and state management across complex multi-vendor networks.

Senior Systems Software Engineer - I May 2024 – Apr. 2026
Paraqum Technologies Pvt. Ltd. *Sri Lanka*

- **Standalone Device Monitoring Platform:** Adapted and scaled the SNMP mapping module into a standalone device monitoring platform, extending real-time health and resource tracking beyond WISP environments.
- **Automated Multi-Protocol Discovery Engine:** Implemented an automated device discovery engine in C++ supporting LLDP, CDP, and vendor-specific protocols for real-time network visibility.
- **API Management System Architecture:** Directed the design and implementation of the Node.js API management system for the Paraqum Network Analyzer.

Electronics Engineer / Associate Electronics Engineer Jun. 2021 – May 2024
Paraqum Technologies Pvt. Ltd. *Sri Lanka*

- **WISP Bandwidth Telemetry:** Developed an SNMP mapping module for the Paraqum WIDI platform, enabling real-time bandwidth management and resource usage monitoring for wireless ISP customers.
- **High-Speed Packet Processing:** Developed C/C++ backend components for the deep packet inspection (DPI) engine to optimize high-speed packet processing.
- **ICMP Device Monitoring:** Implemented real-time device discovery and reachability monitoring services using ICMP-based polling protocols.

Trainee Electronics Engineer Jun. 2019 – Dec. 2019

Paraqum Technologies Pvt. Ltd. *Sri Lanka*

- **Packet Deduplicator (C++):** Engineered a multi-interface packet deduplication system in C++ achieving 8.6 Gbps throughput for real-time session identification across network interfaces.
- **HEVC Video Decoder (C++):** Implemented the weighted prediction feature in a C++ HEVC decoder to improve video decoding performance and spec compliance.
- **I2C Core Hardware Wrapper (Verilog/Vivado):** Designed an RTL wrapper in Verilog to communicate with a clock multiplier module and automate predetermined data transfers via I2C.

Education

University of Moratuwa Sri Lanka

Honours Degree of Bachelor of Science of Engineering *2016 – 2021*

- Academic Standing – **First Class**
- Major – Electronic and Telecommunication Engineering
- Overall GPA – **3.72 / 4.20**

President's College, Embilipitiya Sri Lanka

GCE Advanced Level (2015) in Physical Sciences & GCE Ordinary Level (2012) *2002 – 2015*

- G.C.E. A/L: Ranked **97th** in Sri Lanka (out of 32,393 students) with a Z-Score of 2.5327 | Ranked **2nd** in Ratnapura District
- G.C.E. O/L: 9 A passes | Awarded Dialog Merit Scholarship for outstanding national performance

Skills

Programming: C, C++, Python, Node.js, Java, Assembly, MySQL
Systems Software: Multi-threading, Multi-processing, Linux, Object-Oriented Programming
Developer Tools: Git, Jenkins, MATLAB
Leadership & PM: Technical Leadership, Product Management, Team Mentoring, Agile/Scrum

Leadership & Achievements

- Team Leader – AIESEC (2018)
- Organizing Committee Member – Sri Lanka Robotics Challenge (2018)
- Organizing Committee Member – EXPOSE Exhibition, University of Moratuwa (2018)
- Bronze Medal – Sri Lanka Physics Olympiad (2016)
- Dialog Merit Scholarship issued by Dialog Axiata PLC and Ministry of Education Sri Lanka (2014)

Languages

English – Fluent
Sinhalese – Native

Referees

Mr. P. M. K. Tharaka
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